Status	Finished
Started	Sunday, 18 October 2026, 9:55 PM
Completed	Sunday, 18 October 2026, 9:58 PM
Duration	3 mins 19 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. The parts are merged without sorting
c. Each part is sorted independently
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Merging
b. Rearranging
c. Searching
d. Sorting

The correct answer is: Sorting

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **5**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sort()
b. sorted()
c. arrange()
d. order()

The correct answer is: sorted()

Question **6**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By using the bubble-up method
- c. By sorting data in a single pass
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It increases the number of comparisons needed
c. It allows for faster searching
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Duplicates
c. Selection
d. Searching

The correct answer is: All of the above

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It allows for efficient parallel processing and sorting of data
b. It simplifies the sorting process by using only one pass
c. It avoids the need for recursion
d. It only works on small datasets

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Searching for data in a list
c. Arranging data in a particular format
d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Insertion Sort
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It divides the list into two halves, sorts each half, and then merges them
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It compares adjacent elements and swaps them if necessary
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort